

Incident Reporting System

Project Report

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- **Mr. Md. Tarek Khan** - For his guidance on penetration testing techniques and security best practices.

Their expertise, patience, and constructive feedback have been crucial in helping me navigate complex technical challenges and develop a comprehensive understanding of the DevSecOps ecosystem.

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1 Executive Summary

This capstone project demonstrates the complete DevOps and Security lifecycle of a modern web application. The primary objective was to build, deploy, automate, and secure a Java-based web application using industry-standard tools.

The project successfully integrates software development, Continuous Integration/Continuous Deployment (CI/CD) pipelines, container orchestration via Kubernetes, GitOps principles using ArgoCD, and comprehensive security testing (VAPT). This report details the architecture, implementation phases, and security findings derived from simulating a real-world enterprise environment.

2 Project Architecture & Technologies

The solution was designed to ensure scalability, automation, and security. The following technologies were utilized to achieve the project goals:

- **Application Layer:** Java (Core/Spring Boot), JDBC, PostgreSQL.
- **Infrastructure & OS:** Linux Server Management, Docker Containerization.
- **CI/CD & GitOps:** Jenkins (CI), Git/GitHub (VCS), ArgoCD (GitOps).
- **Orchestration:** Kubernetes (K0s).
- **Security (VAPT):** Nessus (Vulnerability Assessment), Burp Suite/Nmap (Penetration Testing).

3 Implementation Phases

3.1 Phase 1: Application Development

A Java-based web application was developed to interact with a database, featuring user registration, login capabilities, and record management.

- **Key Features:** Basic input validation, exception handling, and file I/O for logging.
- **Structure:** The project adheres to a standard structure including `src/`, `pom.xml`, `Dockerfile`, and `Jenkinsfile`.

3.2 Phase 2: Linux Fundamentals & Source Control

The development environment was configured on a Linux server. This involved user management, package installation (Java, Maven, Git), and initializing version control. The application source code was pushed to GitHub to facilitate collaboration.

3.3 Phase 3: Continuous Integration (CI) with Jenkins

A robust CI pipeline was established using Jenkins to automate the build process. The pipeline consists of the following stages:

1. **Checkout:** Retrieving source code from GitHub.

2. **Build & Test:** Compiling with Maven and running unit tests.
3. **Containerization:** Building a Docker image and pushing it to the container registry.

3.4 Phase 4: Containerization & Kubernetes Deployment

The application was containerized using Docker and deployed to a Kubernetes cluster.

- **Namespace:** A dedicated namespace `ba-cirt-infra` was created for isolation.
- **Service Configuration:** Kubernetes Services and Ingress were configured to expose the application.
- **Domain Mapping:** The domain `ba-industrial-training.com` was mapped to the Kubernetes master node IP.

3.5 Phase 5: GitOps Deployment with ArgoCD

To implement GitOps principles, a separate repository was created for Kubernetes manifests. ArgoCD was configured to monitor this repository and automatically sync changes to the cluster, ensuring the live state matches the desired state defined in Git.

4 Security Assessment (DevSecOps)

4.1 Phase 7: Vulnerability Assessment (VA)

A comprehensive vulnerability scan was performed on the infrastructure using **Nessus**.

- **Scope:** Identification of misconfigurations and network vulnerabilities.
- **Outcome:** A report detailing risk ratings and CVSS scores was generated (See Deliverables).

4.2 Phase 8: Penetration Testing (PT)

Manual penetration testing was conducted focusing on the **OWASP Top 10** vulnerabilities using Burp Suite and Nmap. The testing focused on authentication/authorization testing and configuration weaknesses.

5 Project Deliverables

The following artifacts confirm the successful completion of the project requirements:

1. **Source Code:** [Code- Click Here](#)
2. **Jenkins Pipeline:** [Jenkinsfile- click here](#)
3. **Docker Configuration:** [Dockerfile- click here](#)
4. **Kubernetes Manifests:** [Manifest repo- Click here](#)

5. ArgoCD Deployment Evidence: Accessing ArgoCD Web Interface.

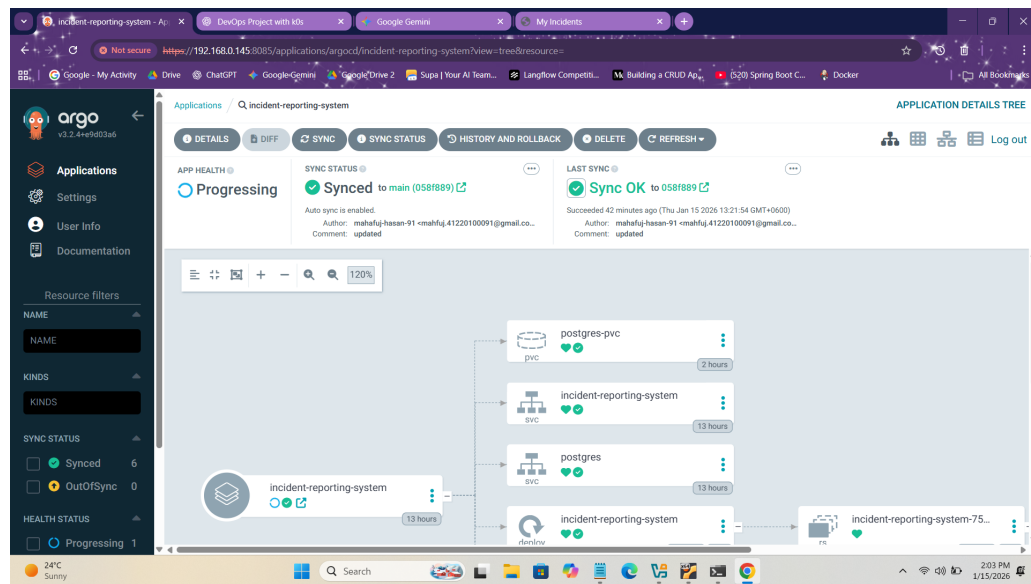


Figure 1: ArgoCD Sync Status

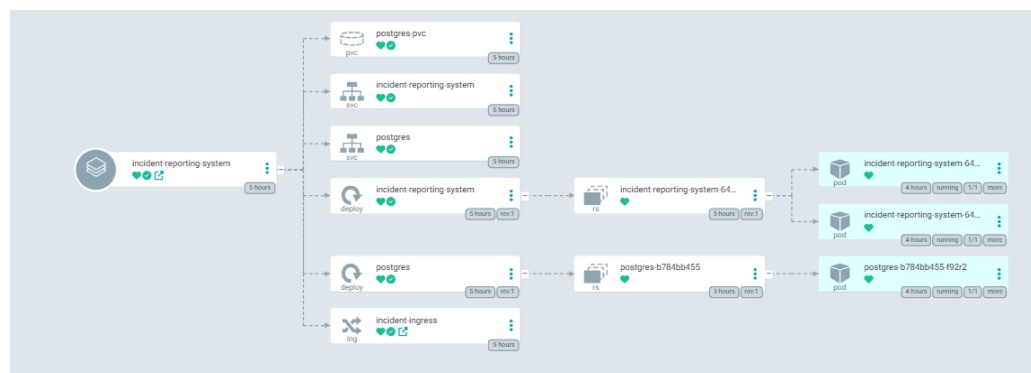


Figure 2: Deployment Architecture and Status

6. Application Access Evidence: Accessing the webpage using ba-industrial-training.com.

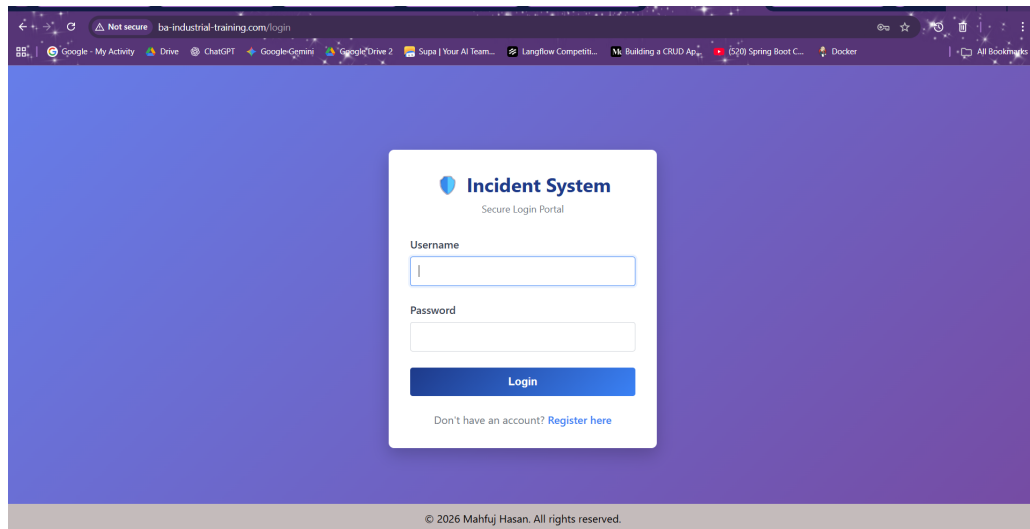


Figure 3: Accessing ba-industrial-training.com

7. **Security Reports:** Attached separately (Nessus Scan & PT Report).

6 Conclusion & Skills Acquired

This project successfully validated the end-to-end flow of a DevSecOps pipeline. By integrating automated deployments with rigorous security testing, the project ensures that the application is not only functional but also secure.

Key Skills Gained

- Java Backend Development & Linux Administration.
- Building CI/CD Pipelines & Docker/Kubernetes Orchestration.
- GitOps Deployment strategies using ArgoCD.
- Vulnerability Assessment & Ethical Hacking methodologies.